

Chia-Hsuan Chao

Computational design, fabrication and interactive systems.

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Professional experience

2024

Gramazio Kohler Research, ETH Zürich

Research Assistant

- Contributed to immersive visualization and interaction software, including tracking, networked interaction and virtual-environment materials.

2021–2022

PKD Engineering Consultants

Computational design & BIM

- Worked on geometric rationalisation, 3D modelling and the translation of design information into fabrication drawings.

2018–2019

Maker-space facilities

Technical support

- Supported laser-cutting and 3D-printing equipment through facilities maintenance and operation.

July 2017

UrbanWASABI Architect

Architectural design contribution

- Contributed to competition design development and 3D modelling.
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Education

2023–2024

ETH Zürich

Master of Advanced Studies in Architecture and Digital Fabrication

Programme study and collaborative research in computational design, robotic fabrication and extended reality.

2017–2020

National Cheng Kung University

Architecture — Design, Computing and Digital Fabrication

Graduate study; bending-active metal-panel thesis research.

2012–2017

Fu Jen Catholic University

Landscape Architecture

Undergraduate study in landscape architecture.

Skills and practice

Geometry to fabrication

Rhino / Grasshopper / Python / COMPAS

Interactive design tools

Unity / C# / Motion capture / MQTT / Firebase

Robotic workflows

COMPAS / ROS / MoveIt / JSON

Generative form

GHPython / Rhino.Geometry / Mola / Grasshopper

Research and publications

2023 · CAADRIA 2023 proceedings

Bending-Active Metal Panel Deformation – Control Through Computational Simulation and Pattern Development

Chia-Hsuan Chao, Kane Yanagawa

Awards

2016

National Student Landscape Design Competition

First prize

2017

National Landscape Architecture New Generation Award

First prize

Project contributions, collaborators and methods are documented at JoshHsuan.github.io.